SS-31 (Mitochondrial Peptide)

Step-by-Step Reconstitution Process

- **Gather supplies: BAC water, alcohol swabs, insulin syringe, vial**
- Bring BAC water and peptide vial to room temperature before mixing
- **Removing plastic cap.** Carefully pop off the plastic cap, do not remove metal flange.
- **Clean both vials:** Wipe the rubber stoppers with alcohol swabs
- **Draw BAC water:** Pull your chosen amount into the syringe
- **Inject slowly:** Add the water to the peptide vial, aiming it at the glass wall—not directly onto the powder
- **Swirl gently:** Roll the vial between your palms. Never shake.
- **Wait:** Let it sit 5-10 minutes. Solution should be crystal clear.
- **Store:** Refrigerate immediately. Good for 28-30 days once reconstituted.

If it's cloudy or won't dissolve: Let it sit longer. Try gentle warming (under your arm or in lukewarm water). If still cloudy after 30 minutes, contact your source.

Dosing Protocol

Clinical Doses

The Barth syndrome trials used 40 mg daily subcutaneously. Those patients have severe mitochondrial dysfunction from a genetic condition. That dosing is for people with fundamentally broken mitochondria, not age-related decline.

The mouse studies used doses that translate to roughly 5 to 7 mg daily in humans.

Practical Protocol for Age-Related Decline

For otherwise healthy individuals experiencing age-related mitochondrial decline:

None

REPAIR PROTOCOL

Dose: 1 to 2 mg per day

Frequency: Once daily

Timing: No fasting required, no specific timing constraints

Administration: Subcutaneous injection

Duration: 4 to 8 weeks (repair phase)

None

RECONSTITUTION (10 mg vial)

Bac water: 2 mL

Concentration: 500 mcg per 10 units on insulin syringe

1 mg per 20 units on insulin syringe

Protocol Sequencing (Repair First Approach)

For people over 35 to 55 with declining energy, slower recovery, and metabolic issues:

Weeks 1 through 4 (or up to 8): Repair Phase SS-31 at 1 to 2 mg daily plus NAD+ at 100 mg subcutaneously three times per week. You are repairing accumulated mitochondrial damage while giving your cells the extra fuel they need to optimize energy.

Weeks 5 through 12: Optimize Phase Replace SS-31 with MOTS-c. Start with 10 mg split into three injections per week (Monday, Wednesday, Friday). Continue NAD+ at 100 mg three times per week. Now you are optimizing the repaired mitochondria.

After the 12-week cycle: Take 4 to 8 weeks completely off. Then you can repeat this protocol 2 to 3 times per year.

Prevention First Protocol

For younger or healthier individuals who want preventative optimization:

Start with MOTS-c at 10 mg split into three injections per week (Monday, Wednesday, Friday) plus NAD+ at 100 mg three times per week. At weeks 4 through 8, assess whether you still have significant fatigue, poor recovery, or metabolic issues. If yes, add SS-31 at 1 to 2 mg daily for 4 to 8 weeks while continuing MOTS-c and NAD+. If not, continue with just MOTS-c and NAD+ for maintenance.

A Note on Higher Doses

You may see dosages recommended online as high as 5 mg, 10 mg, or even 50 mg per day. Those protocols are for people with diseases like Barth syndrome or other conditions involving systemic mitochondrial dysfunction. If you are in your 40s with declining energy and slow recovery, you do not have Barth syndrome. You have accumulated damage from decades of life. Lower doses work for repair because your mitochondria are damaged, not fundamentally broken.

Common Questions

Why should I start with SS-31 instead of MOTS-c? If you have significant mitochondrial dysfunction, SS-31 first is still the right call. MOTS-c tells your cells to produce more energy (it is the supercharger). SS-31 repairs the structural damage (it is the mechanic). Revving a broken engine harder just creates more exhaust. Fix the structure first, then optimize. But if you have moderate age-related decline and are otherwise active and functional, newer research shows these two compounds reduce ROS through completely non-overlapping mechanisms, membrane level for SS-31 and nuclear transcription level for MOTS-c, which means running them concurrently is a viable option. For younger, metabolically healthy people, starting with MOTS-c alone may be appropriate because they may not have significant structural damage yet.

How quickly does SS-31 work? It reaches peak concentration within 15 minutes and has a half-life of about 2 hours. But mitochondrial repair takes time. Most people notice subtle energy and recovery improvements within 1 to 2 weeks, with the full repair phase playing out over 4 to 8 weeks.

Why are the injection site reactions so common? SS-31 can trigger mild histamine responses at the injection site. This is the most frequently reported side effect across all clinical trials. Rotating injection sites helps, and most reactions are mild. Antihistamines or hydrocortisone cream can manage them if needed.

Can I just take NAD+ without SS-31? You can, but if your mitochondria are damaged, you may not get much benefit. Converting NAD+ precursors into usable NAD+ requires ATP. If your mitochondria cannot produce ATP efficiently, they cannot use NAD+ precursors. This is exactly why some people supplement with NAD+ and do not feel anything. The machinery is too broken to use the fuel.

Is SS-31 cleared through the liver? No. SS-31 is completely cleared through the kidneys with no processing through the liver. If you have kidney issues, talk to your doctor before considering this.

Side Effects

In Published Research

Across all clinical trials (over 3 years of daily use in Barth syndrome patients), the safety profile is strong. No serious drug-related adverse events have been consistently reported.

Common:

- Erythema (redness) at injection site: 57%
- Pruritus (itching): 47%
- Pain at injection site: 20%
- Urticaria (hives): 20%
- Irritation: 10%

Less Common:

- Headache (reported in some trials)
- Fatigue (paradoxically, some users report initial fatigue before improvement)
- Gastrointestinal discomfort (rare)

Most injection site reactions are mild and resolve without intervention. Rotating injection sites helps reduce local reactions.

What Users Report

User reports align closely with clinical trial data. Injection site reactions are the primary complaint, with most users describing them as manageable. Some users report initial fatigue in the first few days that resolves as the repair process progresses.